trade.xyz mark: current rules and proposal

highlighted = differs from the current rules 2026-11-11

	Current (deployed)	**Proposal** (three-vote mark, one formula)
Oracle, feed live	$O_t = \mathcal{D}_t\big(\text{clip}(E_t,\ O_{t-k}(1 \pm c))\big)$	$O_t = \mathcal{D}_t(E_t)$
Published input 1	$X^1_t = \mathcal{D}_t\big(\text{clip}(O_t,\ M_{t-1}(1 \pm c))\big)$	$X^1_t = V_t$, $V_t = E_t$ if the feed is live, else O_t
Published input 2	$X^2_t = \mathcal{D}_t\big(\text{clip}(E_t + e_t,\ \text{hull}[\,O_{t-k}(1 \pm c),\ X^2_{t-k}(1 \pm c)\,])\big)$	$X^2_t = A_t$, $\tilde{A}_t = A_{t-1} + \big(1 - e^{-\Delta t/\tau}\big)(V_t - A_{t-1})$, $A_t = \tilde{A}_t$ if B_{t-1} was strictly between V_{t-1} and A_{t-1}, else $A_t = \text{clip}\big(\tilde{A}_t,\ [V_t, M_{t-1}]\big)$
Mark (Hyperliquid)	$M_t = \text{clip}\big(\text{med}(X^1_t, X^2_t, B_t),\ M_{t-1}(1 \pm 1\%)\big)$	$M_t = \text{clip}\big(\text{med}(V_t,\ A_t,\ B_t),\ M_{t-1}(1 \pm 1\%)\big)$
Feed failure (bad print)	no rule	no separate rule: one bad vote out of three does not move the median
Feed not live	scheduled closures only: $O_t = O_{t-1} + \big(1 - e^{-\Delta t^*/\tau}\big)(\text{mid}_t - O_{t-1})$	closure or $E_t = E_{t-1} = E_{t-2}$: $O_t = O_{t-1} + \big(1 - e^{-\Delta t^*/\tau}\big)\ \text{clip}\big(\text{mid}_t - O_{t-1},\ \pm\kappa\, O_{t-1}\big)$ same mark formula: A_t keeps running, never restarted
Liquidation (Hyperliquid)	below MM: 20% of $\lvert q\rvert$ per round above \$100k, rounds 30 s apart, else all	below MM, above \$100k, at every update: $\Delta q = \left(1 - \frac{\text{equity}}{\text{IM}}\right)\lvert q\rvert$; else all

t: relayer update, about every 3 s, read lag $k \geq 1$ E_t: external price $B_t = \text{med}(\text{bid}_t, \text{ask}_t, \text{last}_t)$: xyz book price, computed by Hyperliquid O_t: oracle M_t: mark
V_t: the feed's vote A_t: anchor, $A_0 = E_0$ $c = 50$ bp $d = 1/\text{maxLev}$ $\kappa = 1\%$ $\tau = 30$ min $\Delta t^* = \min(\Delta t, \tau/10)$ $\mathcal{D}_t(x) = \text{clip}\big(x,\ E_t(1 \pm d)\big)$ e_t: 150 s average of mid $- E$
When the feed is not live the relayer keeps O_t unrounded and publishes it rounded. Feed live: the venue is open and E_t changed in at least one of the last two updates; otherwise the feed is not live $[a, b]$: the range between a and b MM $= \frac{1}{2}$ IM at max leverage. Backstop below $\frac{2}{3}$ MM and ADL are unchanged.

The proposal in words: the mark is the median of three votes, the external feed, xyz's own book, and a 30-minute average of the feed (the anchor). No single vote can move it. The anchor is kept between the feed's vote and the previous mark, so it never lags a move the mark has already made; it does not follow a previous mark that xyz's book alone decided, so a push of the book is not carried on after it ends. The anchor is exempt from the discovery bound $\mathcal{D}_t$: the median always lies between V_t and A_t.